1- CLASS TASKS SOLUTIONS

--- TASK 1, 2, 3 & 4 SOLUTION KEY (PYTHON CODE) ---

```Python

```
import numpy as np

Task 1: Matrix Setup
cluster_revenue = np.array([
    [120.0, 135.0, 128.0], # Cluster A
    [90.0, 85.0, 92.0],   # Cluster B
    [210.0, 225.0, 218.0]   # Cluster C
])

print("Array:\n", cluster_revenue)
print("Shape:", cluster_revenue.shape) # (3, 3)
print("Data Type:", cluster_revenue.dtype) # float64

Task 2: Vectorized Target Variances
monthly_target = 110.0
target_variance = cluster_revenue - monthly_target

monthly_totals = np.sum(cluster_revenue, axis=0) # Down columns
cluster_totals = np.sum(cluster_revenue, axis=1) # Across rows

print("\nTarget Variance:\n", target_variance)
print("Monthly Totals (Jan, Feb, Mar):", monthly_totals) # [420.0, 445.0,
438.0]
print("Cluster Totals (A, B, C):", cluster_totals)   # [383.0, 267.0,
653.0]

Task 3: Dispersion & Boolean Masking
overall_mean = np.mean(cluster_revenue)
overall_std = np.std(cluster_revenue)

below_target_mask = cluster_revenue < monthly_target
failed_values = cluster_revenue[below_target_mask]

print("\nOverall Mean:", round(overall_mean, 2)) # 144.78
print("Overall Std Dev:", round(overall_std, 2)) # 48.98
print("Below Target Mask:\n", below_target_mask)
print("Failed Values:", failed_values)   # [90.0, 85.0, 92.0]
```

**--- TASK 4 SOLUTION KEY (BUSINESS MEMO) ---**

"To: Elena Rostova, Regional Sales Director Re: Regional Store Cluster Performance Audit

An audit of quarterly cluster sales reveals significant sales dispersion across regions (standard deviation of $48.98k against a mean of $144.78k). Cluster B consistently failed to hit the $110.0k monthly target in all three months, generating a total of only $267.0k compared to Cluster C's $653.0k. I recommend investigating operational bottlenecks in Cluster B while examining high-performing strategies in Cluster C."
```

2- HOMEWORK SOLUTION

```Python

**# Homework Problem 1 & 2 Python Solution**
import numpy as np

stockout_matrix = np.array([
    [2, 4, 3, 5, 1, 3],    # North
    [8, 12, 10, 14, 9, 11],# South
    [1, 0, 2, 1, 3, 0],    # East
    [6, 7, 5, 8, 9, 6]     # West
])

**# Problem 1**
shape = stockout_matrix.shape # (4, 6)
wh_totals = np.sum(stockout_matrix, axis=1)  # [18, 64, 7, 41]
monthly_avgs = np.mean(stockout_matrix, axis=0)# [4.25, 5.75, 5.0, 7.0,
5.5, 5.0]
total_stockouts = np.sum(stockout_matrix)     # 130

**# Problem 2**
max_target = 5
stockout_var = stockout_matrix - max_target
overall_std = np.std(stockout_matrix)          # 3.91
breaches = stockout_matrix[stockout_matrix > max_target] # [8, 12, 10, 14,
9, 11, 6, 7, 8, 9, 6]

print("WH Totals:", wh_totals)
print("Monthly Avgs:", monthly_avgs)
print("Total Enterprise Stockouts:", total_stockouts)
print("Overall Std Dev:", round(overall_std, 2))
print("Breached Values:", breaches)
```

Homework Grading Criteria:

- Correctly calculates total stockouts = 130.
- Identifies South Warehouse as the primary failure point (64 stockouts; breached target in all 6 months).
- Notes East Warehouse's high compliance (only 7 total stockouts).
- Explains that the standard deviation (3.91) highlights severe operational variance across locations rather than a uniform supply shortages across all warehouses.